2026 SHENZHEN INTERNATIONAL FINTECH COMPETITION · FINTECHATHON

International Track · Topic E: SME Finance Copilot · Theme “AI as a Financial Participant”

LoanReady

An SME finance copilot that turns messy records into clean books, a 13-week cash plan and a lender-ready application, with every action behind a permission gate written in plain code.

Team Korikath Nexus · Bangladesh

github.com/Mrb-Rafi/loanready

THE TEAM

Korikath Nexus

M. Rafiul Bahar Rafi

Team leader

Bioinformatics Engineering, final year
Bangladesh Agricultural University

M. Wasiul Bahar Wasi

Member

Finance and Banking, second year
Bangladesh University of Professionals

Fawzia Tahsin

Member

Bioinformatics Engineering, final year
Bangladesh Agricultural University

Taposhi Rabeya

Member

Bioinformatics Engineering, final year
Bangladesh Agricultural University

ADVISOR

Dr. Md. Rakib Hassan

Professor, Department of Computer Science and Mathematics, Bangladesh Agricultural University

THE PROBLEM

Small businesses are rarely refused credit because they are bad risks. They are refused because they cannot prove what they earn.

Records everywhere

Cash, paper receipts, chat orders and one account shared by personal and business money

Crunch comes first

A large bill falls due before customers pay, and nobody saw it coming

Fastest money costs most

Merchant cash advances priced as a “factor rate” hide an annualised cost of hundreds of percent

PAIN POINTS AND HOW WE SOLVE THEM

Six pains, six answers

Personal and business money mixed

Personal spending is found, confirmed by the owner and moved to owner’s drawings

No books, receipts in a drawer

POS, bank feed and receipts become balanced double-entry books, three statements and a bank reconciliation

The cash crunch is seen too late

A 13-week forecast in low, base and high cases names the first negative week and its drivers

Expensive emergency credit

Close the gap cheapest first: collect, delay, then borrow, ranked by true annualised cost (EAR)

Cannot prove earnings to a lender

A 0–100 readiness score with steps to improve, and a 13-section lender-ready PDF pack

No time, and no trust in an AI with money

The agent does the work, but nothing leaves without an approval bound to the exact payee and amount

MAJOR GOALS

What LoanReady sets out to do

01

Make the business lender-ready

Turn messy records into correct books, statements a lender can read, an honest readiness score and an application pack.

02

Keep cash positive at the lowest cost

See the gap weeks ahead, close it with the cheapest lever first, and show the true annualised cost of every loan.

03

Let AI act, safely

An agent that really sends reminders, pays suppliers and submits the pack in a sandbox, only after human approval, with every step logged.

THE DEMO STORY · RINA'S BAKERY (SYNTHETIC) · 14 OCTOBER 2026

From a shoebox to a loan application

1 Capture
POS, bank and receipts in; 3 personal costs, 1 duplicate found

2 Books
P&L, cash flow, balance sheet; ledger cash = bank + till

3 Forecast
Cash goes negative in week 3: a 6,500.00 flour order

4 Collect
Reminders for 2 overdue invoices, approved and sent

5 Delay
A penalty-free bill moved within its grace period

6 Borrow
Last 1,000.00: supplier credit at EAR 0.00% beats an MCA

7 Pay
The regular flour supplier paid after approval

8 Defend
Fake bank-account bill blocked; planted instructions ignored

9 Loan pack
Readiness 72 → 86; pack submitted, receipt back

10 Evidence
Action log exported, hash chain verified intact

All data synthetic; all money moves inside a self-built sandbox bank, POS, messenger and lender

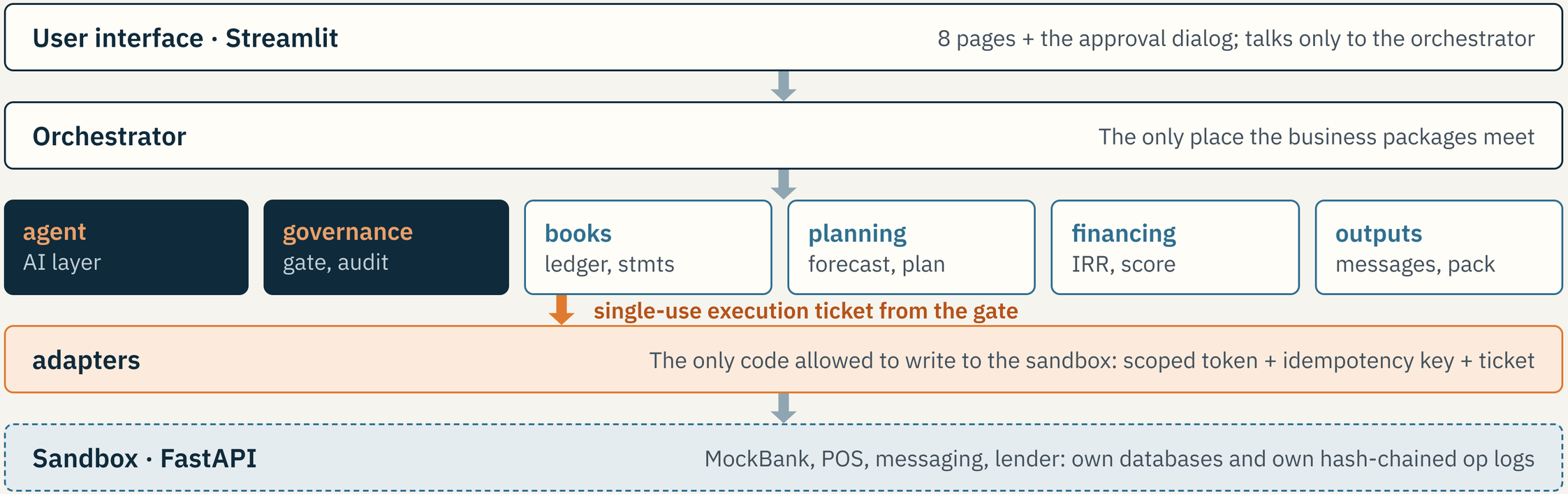
THE ONE RULE BEHIND THE WHOLE DESIGN

AI proposes. Code decides.

The model never computes money, never checks permissions and never chooses a payee, amount or account. Those are plain, tested code.

TECHNOLOGY · ARCHITECTURE

One machine, strict layers, one door to the bank



Business packages never import each other: checked by lint-imports on every build

TECHNOLOGY · ENGINEERING RULES

Rules enforced by code and tests, not by convention

Money is exact

Integer cents between modules, Decimal inside, half-even rounding in one place. A static check rejects floats near money.

Deterministic

Fixed business clock, seeded IDs, canonical hashing: the demo and evidence are byte-identical on Windows and Linux.

Works with nothing installed

Mock AI, OCR sidecar files and an in-process sandbox: no model, API key or server needed to run or test.

Tested at every layer

719 tests pass across 13 suites: golden values, property tests, injection corpus, end-to-end demo.

Open stack

Python 3.12 · Streamlit · FastAPI · SQLite · Pydantic · reportlab · Ollama / OpenAI, all permissively licensed.

One-command install

Linux, macOS and Windows installers set up, seed and start the app in a few minutes.

Latest run: 719 passed, 4 skipped (deliberately unbuilt features) · evidence/test_report.txt

FINANCE · BOOKS AND THE CASH PLAN

Correct books first, then close the gap cheapest first

Accrual books, IFRS for SMEs

- Rules first, AI second for categories; personal items to drawings
- Balanced double-entry journals; duplicates flagged
- P&L, direct cash flow, balance sheet: identities checked
- Bank reconciliation: ledger cash = bank + till
- 13-week forecast: low, base and high cases, first negative week and its drivers

Rina’s gap, closed in cost order

1 Collect overdue invoices	3,200.00
2 Delay a penalty-free bill	600.00
3 Borrow only the rest	1,000.00

Gap to close 4,800.00 (USD, synthetic). Collect and delay cost nothing; borrowing comes last.

Accounting policies are written into the loan pack (IFRS for SMEs Section 8)

FINANCE · THE TRUE COST OF CREDIT

Every option ranked by annualised cost, neutrally

Option for 1,000.00 over 49 days	Total cost	Est. APR	EAR
Supplier 30-day credit (cheapest)	0.00	0.00%	0.00%
Term loan, 12 months	76.19	14.03%	15.06%
Business overdraft	24.16	17.79%	19.46%
Supplier 2/10 net 30, discount lost	20.00	36.89%	44.59%
App micro-loan, 30 days	50.00	59.41%	81.05%
Merchant cash advance (most expensive)	200.00	293.10%	**1,752.76%**

Fictional products dated 2026-10-14; not offers. Ranked by EAR; no commissions, no sponsored order.

How the cost is computed

```
NPV(r) = Σ cf / (1+r)^day
daily IRR r by bisection
APR = r × 365
EAR = (1+r)^365 - 1
```

Exact cash flows in cents, Decimal at 34-digit precision, checked against golden values. A “factor rate” of 1.20 looks like 20%; its true annualised cost is shown beside it with a plain explanation.

Annualised rate displayed prominently, aligned with PBoC Announcement [2021] No. 3 and US commercial-financing disclosure laws

FINANCE · LOAN READINESS

A score the owner can improve, and a pack a lender can read

Component	Points	Measured from
Bookkeeping	20	categorised and reconciled share
History	15	months of records
Separation	15	personal spending separated
Stability	15	variation of weekly sales
Cash health	15	weeks with positive operating cash
Receivables	10	overdue share of what is owed
Debt service	10	debt-service coverage ratio

Business data only: the input has no personal attributes. A self-assessment, not a credit score.

72 → 86

Rina, FAIR to GOOD after her confirmations

Lender-ready PDF pack

- 13 fixed sections: statements, forecast, reconciliation, ageing, readiness, financing, policies, controls
- Content hash on every page; the file’s SHA-256 is re-checked by the gate and by the lender
- Submitted only after approval; accepting a loan is always blocked

Improvement steps are computed by re-scoring the improved input, not guessed

SECURITY · PERMISSION TIERS

What the agent may do, fixed in code

T0 Observe

Runs at once

Categorise, build statements, reconcile, forecast, plan, compare financing, score readiness

T1 Propose

Drafts only

Draft reminders, delay requests and the loan pack; nothing leaves LoanReady

T2 Approve

Human approval, exact details

Send reminders, pay a known supplier, cancel a payment, submit the loan pack

Blocked

Never executes

Pay a new payee, change a payee’s account, accept a loan, raise limits, disable the gate or kill switch, edit the log

Maker-checker: nobody approves their own proposal; a bookkeeper cannot approve payments.

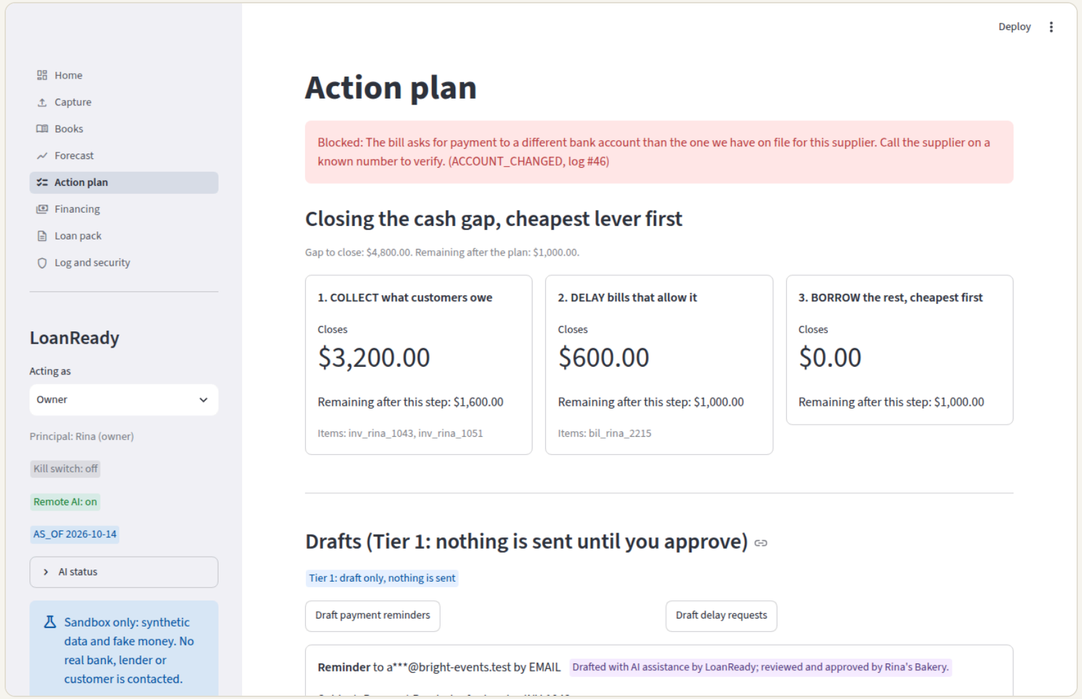
Caps and kill switch: 5,000.00 per action, 10,000.00 per day; one switch stops every send and payment.

The tier table is read-only with no setter; unknown action types fail closed · docs/generated/tier_table.md

SECURITY · THE PERMISSION GATE

Eight checks before anything happens

1. Kill switch
2. Blocked set or unknown action type
3. Origin: document text cannot start an action
4. Role of the person asking
5. Parameter schema
6. Entities vs stored records: **NEW_PAYEE, ACCOUNT_CHANGED**
7. Caps and message rate limits
8. Tier decision: allow, draft or wait for approval

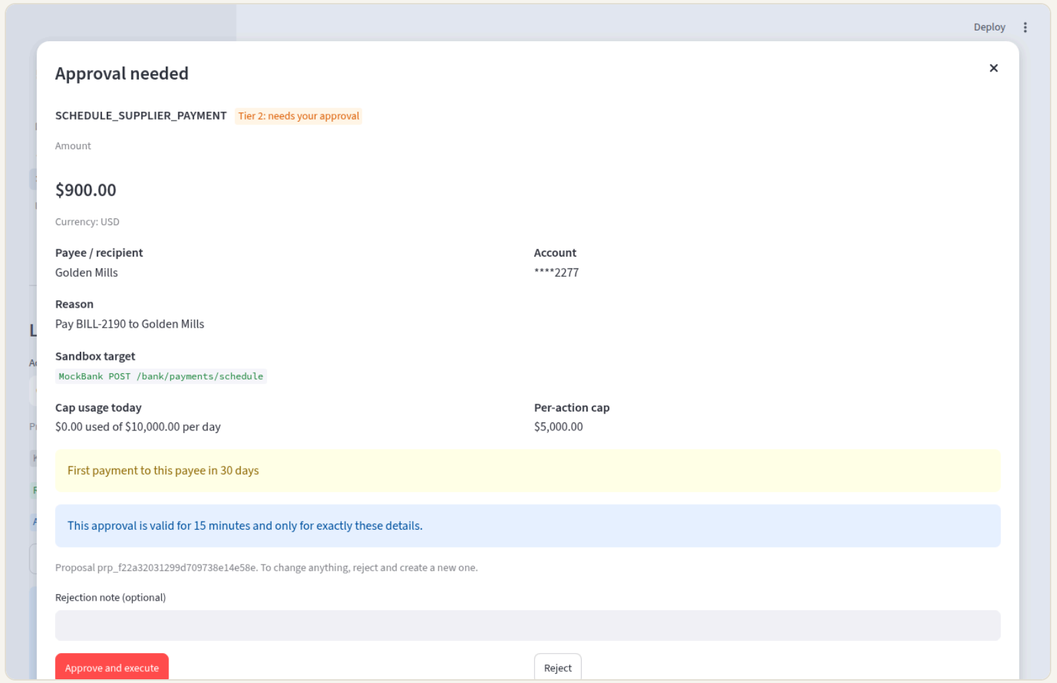


A bill asking to be paid to a new bank account is blocked and logged; the gate checks again at execution, and MockBank checks a third time.

The first failed check stops the request and writes a BLOCKED row to the audit log

SECURITY · APPROVALS AND AUDIT

Approve once, exactly, with proof



The owner sees the exact payee, amount and account before approving.

Approval token

Bound to the exact parameters (PSD2 dynamic linking); single use; 15 minutes.

Execution ticket

HMAC-signed, single use; the sandbox refuses any write without one.

Hash-chained audit log

Every decision, blocked too: row_hash = sha256(prev + row). No edit path.

Two independent records

The sandbox bank keeps its own chain; both verify intact.

Replay, expired or changed approvals fail: APPROVAL_REPLAY, APPROVAL_EXPIRED, APPROVAL_PARAMS_CHANGED

COMPLIANCE · REGULATIONS AND LAWS WE DESIGN AROUND

48 rules mapped to 57 tested controls

Financial reporting

IFRS for SMEs (default) · US GAAP option

Accrual books, three statements, going concern, policies note

Internal control

SOX · COSO 2013 · PCAOB AS 2201

Maker-checker, reconciliation, tamper-evident audit trail

Payments

PSD2 SCA · Confirmation of Payee · FATF

Approval bound to amount and payee; changed accounts blocked

Lending and disclosure

PBoC Ann. [2021] No. 3 · CA and NY disclosure laws · EU CCD

Annualised rate shown; compare only, never broker a loan

Data protection

China PIPL · GDPR incl. Chapter V transfers

Data minimisation; bank identifiers and documents stay local

AI governance

EU AI Act · NIST AI RMF · OWASP LLM Top 10 · China AI labelling

Human oversight of every action; AI text labelled

Designed around, not certified under, these regimes · full map in docs/compliance_matrix.md

THE AI LAYER · ITS JOB

AI used narrowly, where it helps

The model may only

- ✓ Classify into a fixed label list
- ✓ Word text from facts computed by code
- ✓ Pick one item from a fixed menu

It never

computes money · checks permissions · chooses a payee, amount or account · follows instructions found in a document

Class	Data	Remote model?
C0 / C1	Public, aggregates	Allowed
C2	Counterparty names	Masked only
C3	Bank identifiers	Never
C4	Receipts, documents	Never

Four modes, one switch

mock (default, tests and demo) · **ollama** local model · **openai** · **hybrid**: messages to OpenAI, receipts and intents stay local, mock always last

Data-class checks are plain code in agent/policy.py, enforced before any network call

THE AI LAYER · EVERY CALL IS GUARDED

One pipeline for every model call

Policy
refuse data above the task's class

→

Prompt
untrusted text only inside <data> blocks

→

Route
backends in order, mock last

→

Mask
names, numbers, e-mails out before remote

→

Validate
strict JSON schema; one retry, then next

→

Post-check
every number must match the facts

Injection-proof by design

Planted instructions in a bank memo, POS note, receipt and bill are ignored, flagged and logged; zero actions result.

Never over-trusted

Local-model labels capped at 0.75 confidence, below the 0.80 auto-post line: they pre-fill, the owner confirms.

Labelled and budgeted

AI text is marked as AI-generated. OpenAI spend reserves the worst case against daily and all-time caps.

A message that drops the amount, adds a link or uses threats is rejected and the template is used instead

EVIDENCE · RESULTS FROM THE REAL SYSTEM

Real tasks done, safely, and provable

10/10

demo scenarios S1–S10 pass, checked by one command

719

automated tests passed across 13 suites

5 + 2

actions executed in the sandbox after approval; unsafe ones blocked

47

hash-chained audit rows; both chains verify intact

Submitted to the organisers

Technical documentation · this deck · source code with README and tests on GitHub · security self-assessment · execution evidence (sandbox logs)

Try it in one line (Linux / macOS)

```
curl -fsSL https://raw.githubusercontent.com/ Mrb-Rafi/loanready/main/install.sh | bash
```

Windows: install.ps1 in the same repository

Sources: evidence/demo_summary.json, evidence/action_log.csv, evidence/test_report.txt

LOANREADY · TOPIC E: SME FINANCE COPILOT

From messy records to lender-ready, safely.

Thank you. We welcome your questions.

Team Korikath Nexus

M. Rafiul Bahar Rafi · M. Wasiul Bahar Wasi · Fawzia Tahsin · Taposhi Rabeya

Advisor: Dr. Md. Rakib Hassan

github.com/Mrb-Rafi/loanready